

AUGUSTINE ISLAND, AK, USA

WMO: 994700

Lat: 59.378N

Lon: 153.348W

Elev: 9

StdP: 101.22

Time Zone: -9.00 (NAK)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.6	(c) -15.0	(d) -22.3	(e) 0.5	(f) -14.2	(g) -20.3	(h) 0.6	(i) -12.4	(j) 25.3	(k) -5.2	(l) 23.2	(m) -5.5	(n) 13.1	(o) 270	(p) 0.894

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 3.6	(c) 20.6	(d) 15.3	(e) 19.0	(f) 14.6	(g) 17.4	(h) 13.5	(i) 16.5	(j) 19.8	(k) 15.5	(l) 18.7	(m) 14.5	(n) 17.7	(o) 4.9	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.0	10.7	17.9	13.9	9.9	17.0	12.8	9.2	15.4	45.9	19.6	43.1	18.8	40.3	18.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 21.6	(b) 19.1	(c) 16.9	(e) -20.6	(f) 23.1	(g) 2.8	(h) 2.0	(i) -22.6	(j) 24.5	(k) -24.3	(l) 25.7	(m) -25.9	(n) 26.9	(o) -27.9	(p) 28.4		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-3.3	-2.1	-2.2	1.9	7.1	10.2	12.7	13.0	9.7	5.2	-0.2	-1.9
	DBStd	7.08	5.66	4.41	4.63	3.50	2.90	2.82	2.88	2.36	2.19	2.87	4.81	5.11
	HDD10.0	2357	412	340	377	243	100	29	4	1	30	148	305	369
	HDD18.3	5156	670	573	635	493	348	245	177	167	260	406	555	627
	CDD10.0	248	0	0	0	1	11	35	88	93	20	0	0	0
	CDD18.3	4	0	0	0	0	0	1	2	1	0	0	0	0
	CDH23.3	3	0	0	0	0	0	1	2	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.2	9.0	8.9	8.2	6.8	5.2	4.6	5.2	5.5	6.8	7.9	8.7	9.5
Precipitation	PrecAvg	1343	130	111	86	91	88	74	88	102	148	150	129	146
	PrecMax	1624	260	195	158	146	156	113	129	161	262	269	284	236
	PrecMin	1007	42	48	19	32	34	22	44	33	58	70	33	61
	PrecStd	141	50	37	35	29	34	24	23	34	44	47	60	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	5.3	5.9	12.7	19.8	22.5	23.3	21.4	15.7	10.7	8.1	6.7
		MCWB	3.2	4.5	4.9	6.7	14.2	17.9	15.7	16.2	11.8	8.1	6.8	5.9
	2%	DB	4.6	4.2	4.7	9.7	15.6	19.3	20.9	19.9	14.2	9.8	7.3	5.6
		MCWB	3.5	3.4	3.9	5.3	9.1	14.6	14.8	15.5	10.7	8.1	6.6	5.0
	5%	DB	3.8	3.4	3.9	7.5	13.2	16.9	19.3	18.3	13.1	9.1	6.5	4.7
		MCWB	2.9	2.6	3.0	3.9	7.5	13.1	14.5	14.3	10.2	7.6	5.8	4.1
	10%	DB	2.9	2.6	3.0	5.6	11.4	14.6	17.6	16.8	12.3	8.5	5.7	3.8
		MCWB	2.0	1.8	1.8	3.5	7.0	11.6	13.6	13.1	10.1	7.2	5.0	3.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	5.2	5.9	7.8	12.6	18.0	17.4	17.6	13.4	10.0	7.8	6.5
		MCDB	4.8	5.3	6.4	13.7	18.8	21.2	21.3	20.2	14.5	10.2	7.9	6.6
	2%	WB	4.0	4.3	4.9	6.2	9.4	15.8	16.3	16.4	12.3	8.9	7.0	5.4
		MCDB	4.2	4.7	5.0	10.0	13.0	17.7	20.2	19.6	12.9	9.3	7.2	5.7
	5%	WB	3.2	3.3	3.9	5.2	8.6	14.1	15.4	15.4	11.5	8.3	6.3	4.5
		MCDB	3.5	3.8	4.0	7.1	11.6	16.7	19.3	18.5	12.1	8.8	6.7	4.9
	10%	WB	2.4	2.4	2.6	4.4	8.0	12.7	14.5	14.2	10.8	7.7	5.5	3.4
		MCDB	2.9	2.9	3.0	6.1	10.3	14.8	18.3	17.1	11.5	8.2	6.0	3.9
Mean Daily Temperature Range	5% DB	MDBR	3.8	3.6	3.7	3.4	3.7	4.1	4.1	3.6	2.9	2.8	3.6	3.8
		MCDBR	3.3	3.4	3.4	5.1	6.8	7.7	7.3	5.8	3.9	2.8	3.0	3.8
		MCWBR	2.8	2.9	2.8	2.5	3.7	4.4	3.8	3.1	2.9	2.5	2.6	3.4
	5% WB	MCDBR	3.3	3.0	2.8	3.9	6.0	7.3	7.3	5.9	3.2	2.5	2.7	3.4
		MCWBR	2.9	2.7	2.6	2.5	3.5	4.8	4.1	3.6	2.6	2.6	2.7	3.3
Clear-Sky Solar Irradiance	taub	0.257	0.306	0.319	0.354	0.374	0.366	0.373	0.361	0.338	0.315	0.277	0.270	
	taud	2.251	2.315	2.335	2.309	2.321	2.430	2.455	2.481	2.510	2.414	2.175	2.022	
	Ebn at Noon	563	710	819	848	854	868	852	836	799	695	499	347	
	Edh at Noon	41	67	87	107	114	104	99	90	75	60	43	35	
All-Sky Solar Radiation	RadAvg	0.29	0.76	2.07	3.41	4.84	5.40	4.73	3.78	2.47	1.17	0.43	0.18	
	RadStd	0.04	0.14	0.38	0.51	0.55	0.52	0.47	0.51	0.36	0.13	0.08	0.02	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.80	+2.16	+2.08	N/A	N/A	N/A	N/A	-225	-287	N/A	N/A

Nomenclature: See separate page